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PART 1

SECTION B – API ENDPOINT PLAN

The RaceDay API provides the communication layer between the RaceDay web application and its database. The API allows users to register and log in, manage their profiles, browse events, enrol in event categories, view routes and weather information, and track race results.

Event Organizers are provided with additional functionality that allows them to create and manage events, categories, routes, enrolments and participant results.

The endpoint plan below was created before the implementation of the API and corresponds with the entities identified in the RaceDay Entity Relationship Diagram (ERD).

Role Definitions

The RaceDay system uses role-based access control to determine which users may access particular endpoints.

None (Public)	No authentication is required. The endpoint can be accessed by any user.
Any (Logged in)	The user must be authenticated, regardless of whether they are a Participant or Organizer.
Participant	The user must be logged in with a Participant account.
Organizer	The user must be logged in with an Organizer account.

API Endpoint Plan

Authentication

These endpoints are responsible for creating user accounts and authenticating existing users.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Creates a new RaceDay user account. The user selects either the Participant or Organizer role so that the correct profile and permissions can be assigned.	None (Public)	{ fullName, email, password, role }	201 Created – user account created successfully. 400 Bad Request – required or invalid data. 409 Conflict – email address is already registered.
POST	/api/auth/login	Authenticates an existing user using their email and password and returns an authentication token for accessing protected endpoints.	None (Public)	{ email, password }	200 OK – authentication token and user information returned. 400 Bad Request – missing login details. 401 Unauthorized – email or password is incorrect.

Users

These endpoints allow authenticated users to view and update their account information.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/{id}	Retrieves the account information of a specific RaceDay user. It is used to display the user's basic account details.	Any (Logged in)	None	200 OK – user details returned. 401 Unauthorized – user is not logged in. 403 Forbidden – user is not allowed to view the requested account. 404 Not Found – user does not exist.
PUT	/api/users/{id}	Updates the account information of an existing user, such as their full name or email address.	Any (Logged in)	{ fullName, email }	200 OK – user details updated. 400 Bad Request – invalid data. 401 Unauthorized – user is not logged in. 403 Forbidden – user cannot update this account. 404 Not Found – user does not exist. 409 Conflict – email

is already in use.

Organizer Profiles

These endpoints allow Event Organizers to access and maintain their organizer profiles.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/organizers/{id}	Retrieves the profile of a specific Event Organizer, including the organization name and contact details.	Organizer	None	200 OK – organizer profile returned. 401 Unauthorized – user is not logged in. 403 Forbidden – Organizer role required. 404 Not Found – organizer profile does not exist.
PUT	/api/organizers/{id}	Updates an Organizer's organization name and contact information so that their RaceDay profile remains current.	Organizer	{ organizationName, contactNumber }	200 OK – organizer profile updated. 400 Bad Request – invalid data. 401 Unauthorized – user is not logged in. 403 Forbidden – Organizer role required or user cannot edit this profile. 404 Not Found – organizer profile does not exist.

Participant Profiles

These endpoints allow Participants to access and maintain their RaceDay profiles.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/participants/{id}	Retrieves a Participant's profile information for use when managing their RaceDay account and	Participant	None	200 OK – participant profile returned. 401 Unauthorized – user is not logged in. 403 Forbidden – Participant role required. 404 Not Found – participant

		event participation.			profile does not exist.
		Updates a Participant's personal race profile, including date of birth and emergency contact information.	Participant	{ dateOfBirth, emergencyContact, emergencyPhone }	200 OK – participant profile updated. 400 Bad Request – invalid data. 401 Unauthorized – user is not logged in. 403 Forbidden – Participant role required or user cannot edit this profile. 404 Not Found – participant profile does not exist.

Events

The Events endpoints allow the public to browse RaceDay events while allowing Organizers to create and manage events.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Retrieves a list of RaceDay events so that users can browse available running, walking and cycling events.	None (Public)	None	200 OK – list of events returned.
GET	/api/events/{id}	Retrieves complete information about a specific event, allowing users to view its date, location, type, description and status.	None (Public)	None	200 OK – event details returned. 404 Not Found – event does not exist.
POST	/api/events	Creates a new running, walking or cycling event and links the event to the logged-in Organizer.	Organizer	{ eventName, eventType, eventDate, location, description, status }	201 Created – event created successfully. 400 Bad Request – invalid or missing event data. 401 Unauthorized – user is not logged in. 403 Forbidden – Organizer role required.

PUT	/api/events/{id}	Updates an existing event so that an Organizer can change information such as the date, location, description or status.	Organizer	{ eventName, eventType, eventDate, location, description, status }	200 OK – event updated successfully. 400 Bad Request – invalid event data. 401 Unauthorized – user is not logged in. 403 Forbidden – user is not permitted to modify the event. 404 Not Found – event does not exist.
DELETE	/api/events/{id}	Removes an existing event that belongs to the logged-in Organizer.	Organizer	None	204 No Content – event deleted successfully. 401 Unauthorized – user is not logged in. 403 Forbidden – user is not permitted to delete the event. 404 Not Found – event does not exist. 409 Conflict – event cannot be deleted because related records prevent deletion.

Event Categories

Event categories represent the different distances or participation options offered for an event, such as a 5 km run or 10 km run.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Retrieves all categories associated with a specific event so that Participants can see the available distances and entry options.	None (Public)	None	200 OK – list of categories returned. 404 Not Found – event does not exist.
POST	/api/events/{eventId}/categories	Creates a new category for an event and links it to the specified	Organizer	{ categoryName, distanceKm, maxParticipants, entryFee }	201 Created – category created successfully. 400 Bad Request – invalid

		event.			category data. 401 Unauthorized – user is not logged in. 403 Forbidden – Organizer role required. 404 Not Found – event does not exist. 200 OK – category updated successfully. 400 Bad Request – invalid category data.
PUT	/api/categories/{id}	Updates the details of an existing event category, including its distance, capacity and entry fee.	Organizer	{ categoryName, distanceKm, maxParticipants, entryFee }	category data. 401 Unauthorized – user is not logged in. 403 Forbidden – user cannot modify the category. 404 Not Found – category does not exist. 204 No Content – category deleted. 401 Unauthorized – user is not logged in. 403 Forbidden – user cannot delete the category. 404 Not Found – category does not exist. 409 Conflict – category has related enrolments
DELETE	/api/categories/{id}	Removes an existing category from an event when the Organizer no longer wants to offer that category.	Organizer	None	

and cannot be deleted.

Event Enrolments

These endpoints allow Participants to enter RaceDay events and manage their registrations.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/enrollments	Retrieves the logged-in Participant's event enrolments so that they can view the events and categories they have entered.	Participant	None	200 OK – list of enrolments returned. 401 Unauthorized – user is not logged in. 403 Forbidden – Participant role required.
POST	/api/enrollments	Creates an enrolment linking a Participant to a selected event category so that the Participant can take part in the event.	Participant	{ participantId, categoryId }	201 Created – enrolment created successfully. 400 Bad Request – invalid enrolment data. 401 Unauthorized – user is not logged in. 403 Forbidden – Participant role required. 404 Not Found – participant or category does not exist. 409 Conflict – Participant is already enrolled or the category has reached capacity.
GET	/api/enrollments/{id}	Retrieves a specific enrolment record, including the Participant's selected category, registration status and race number.	Participant	None	200 OK – enrolment details returned. 401 Unauthorized – user is not logged in. 403 Forbidden – user cannot view this enrolment. 404 Not Found – enrolment does not exist.
DELETE	/api/enrollments/{id}	Cancels an existing event	Participant	None	204 No Content – enrolment cancelled

enrolment
belonging to
the logged-in
Participant.

successfully. 401
Unauthorized – user is not
logged in. 403 Forbidden –
user cannot cancel this
enrolment. 404 Not
Found – enrolment does
not exist. 409
Conflict – enrolment can
no longer be cancelled.

Results

The Results endpoints support RaceDay's performance-tracking functionality. Participants can view their race history while Organizers can record and correct official results.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/results	Retrieves the logged-in Participant's race results so that they can track their personal performance history.	Participant	None	200 OK – results history returned. 401 Unauthorized – user is not logged in. 403 Forbidden – Participant role required.
GET	/api/results/{id}	Retrieves a specific race result, including finish time, finishing position and result status.	Participant	None	200 OK – result details returned. 401 Unauthorized – user is not logged in. 403 Forbidden – user cannot view this result. 404 Not Found – result does not exist.
POST	/api/results	Creates an official race result for an enrolment after an event has taken place and links the result to the Participant's enrolment.	Organizer	{ enrollmentId, finishTime, position, resultStatus }	201 Created – result recorded successfully. 400 Bad Request – invalid result data. 401 Unauthorized – user is not logged in. 403 Forbidden – Organizer role required. 404 Not Found – enrolment does not exist. 409 Conflict – a result already exists for the enrolment.
PUT	/api/results/{id}	Updates an existing race result when an	Organizer	{ finishTime, position, resultStatus	200 OK – result updated successfully. 400 Bad Request – invalid result

Organizer needs
to correct the
finish time,
position or
result status.

data. 401 Unauthorized – user
is not logged in. 403 Forbidden
– Organizer role required. 404
Not Found – result does not
exist.

Routes

The Routes endpoints provide race-day route information for events.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/routes	Retrieves route information for a specific event so that Participants can prepare for the event and understand the course.	None (Public)	None	200 OK – route information returned. 404 Not Found – event or route information does not exist.
POST	/api/events/{eventId}/routes	Creates a route for an event and stores its name, distance, start point, end point and map reference.	Organizer	{ routeName, distanceKm, startPoint, endPoint, mapUrl }	201 Created – route created successfully. 400 Bad Request – invalid route information. 401 Unauthorized – user is not logged in. 403 Forbidden – Organizer role required. 404 Not Found – event does not exist.

Weather Information

Weather information helps Participants prepare for race-day conditions.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/weather	Retrieves weather information for a specific event,	None (Public)	None	200 OK – weather information returned. 404 Not

including
temperature, rain
probability, wind
speed and general
conditions, to help
Participants prepare
for race day.

Found – event or
weather
information does
not exist.

HTTP Status Codes

The RaceDay API uses standard HTTP response codes to indicate whether a request was successful or unsuccessful.

HTTP Status	Meaning	Use in RaceDay
200 OK	Request successful	Used when information is successfully retrieved or updated.
201 Created	Resource successfully created	Used after registration or when creating events, categories, enrolments, routes or results.
204 No Content	Request successful with no response body	Used when a resource is successfully deleted.
400 Bad Request	Invalid request	Used when required information is missing or incorrectly formatted.
401 Unauthorized	Authentication required	Used when a protected endpoint is accessed without valid authentication.
403 Forbidden	Permission denied	Used when the user is authenticated but does not have the required role or ownership.
404 Not Found	Resource does not exist	Used when a requested user, event, category, enrolment, result or other resource cannot be found.
409 Conflict	Request conflicts with existing data/state	Used for duplicate registrations, duplicate email addresses or records that cannot be deleted because they are in use.

The API endpoint plan is designed to correspond with the database entities identified in Section A.

Maintaining this relationship between the API plan and database design helps ensure that the implementation in Part 2 follows the system plan.

